Settle four Unit 1 arguments

19 I

DIRECTIONS · Shade who is right, give evidence, and fix the wrong claim.

ROUND 1

ANA
"3 m right then 3 m back: the distance is zero."

BEN
"Distance is 6 m. Displacement is zero."

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 2

KAI
"8 N pull on a 2 kg cart: a = 8 ÷ 2 = 4 m/s²."

LU
"With 2 N friction, the net force is 6 N, so a = 3 m/s²."

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 3

MAYA
"−2 → −6 m/s is slowing, because the numbers are negative."

RAJ
"The speed grows from 2 to 6 m/s. It is speeding up."

Who's right? _____ · Evidence: _____

Fixed claim: _____

YOUR TURN · WRITE A TRAP

Write a claim about today's idea that sounds right but is wrong. Trade and settle it.

WORKSHOP 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Unit 1's most wanted

These mistakes show up on every test. Hunt them down.

19 Y

WANTED

FOR CRIMES AGAINST PHYSICS

"Use the pull in F = ma."

LAST SEEN: 8 N pull, 2 N friction, 2 kg

THE TRUTH:

EVIDENCE:

REWARD: NET FORCE FIRST

WANTED

FOR CRIMES AGAINST PHYSICS

"The force pair cancels on one cart."

LAST SEEN: two carts pushing apart

THE TRUTH:

EVIDENCE:

REWARD: DIFFERENT OBJECTS

WANTED

FOR CRIMES AGAINST PHYSICS

"A high x–t line means fast."

LAST SEEN: a flat line at 8 m

THE TRUTH:

EVIDENCE:

REWARD: SLOPE, NOT HEIGHT

CAPTURED! · WRITE YOUR OWN WANTED NOTICE

A wrong idea about today's physics I once believed:

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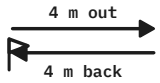
UNIT 1 · DAY 19 · WORKSHOP · SEI · PARTNER TALK

Talk it through: find the mistake

Language goal: I can **explain** a correction.

DIRECTIONS · Partner A and B read their lines aloud. Listen, then write. Switch each round.

ROUND 1 · "4 m OUT AND BACK, SO MY DISPLACEMENT IS 8 m."

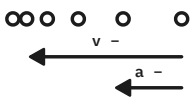


A ASKS "What is wrong with this answer?"

B SAYS "It used ____ instead of ____. The displacement is ____ m."

A CHECKS "So the fix is to compare only ____ and ____."

ROUND 2 · SWITCH · "NEGATIVE a ALWAYS MEANS SLOWING DOWN."

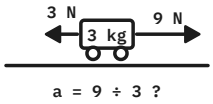


B ASKS "Is that always true?"

A SAYS "No. If v is also ____, the object is ____."

B ADDS "I want to add that we compare the ____ of v and a."

ROUND 3 · BOTH · "9 N RIGHT, 3 N LEFT, 3 kg: a = 9 ÷ 3."



A "The mistake is using ____ instead of the net force."

B "The net force is ____ N, so a = ____ m/s²."

LISTENING LOG · MY PARTNER SAID…

NOW I WRITE IT MYSELF

The most common mistake I found is…

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

WORKSHOP 1 2 3

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19 T

The case of the wrong answer

A returned quiz is the crime scene. Find the first wrong step and fix it.

EXHIBIT A · THE STUDENT'S WORK

Problem: a 3 kg cart has 9 N right and 3 N left. Find a.

Step 1: F = 9 N

Step 2: a = F ÷ m

Step 3: a = 9 ÷ 3

Step 4: a = 3 m/s²

ANALYSIS 1 · FIRST WRONG STEP

Step ____ is wrong because ____

ANALYSIS 2 · REDO IT

EXHIBIT B · YOUR OWN RETURNED WORK

Copy one problem you missed. Circle the first wrong step.

THE RULE THAT CRACKED THE CASE

Name the physics rule and draw the picture that proves it.

THE VERDICT

Next time, I will check ____ before ____ .

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Say it, then write it: fix the reasoning

Language goal: I can **explain a correction.**

mistake	correct	because	evidence
error · erro	corregir · corrigir	porque · porque	evidencia · evidência
a step that is wrong	to fix it	gives the reason	the picture or numbers

English · Spanish · Portuguese. My home language: _____

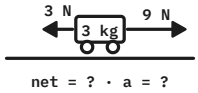
1 · READ AND FILL · WORD BANK: mistake · correct · because · evidence · rule

I found the first ____ in step 2. To ____ it, I used a ____: find the net force first. My ____ is the force diagram. The new answer is right ____ it uses every force.

2 · SAY IT TO A PARTNER, THEN WRITE IT

My original answer treated ____ as ____.
I changed ____ because ____.

3 · SOLVE WITH THE STEPS



net = ? · a = ?

A 3 kg cart: 9 N right, 3 N left. A student wrote a = 9 ÷ 3 = 3 m/s².

STEP 1 · NET FORCE	STEP 2 · DIVIDE	STEP 3 · SAY IT
____ − ____ = ____ N	____ ÷ ____ = ____ m/s²	"The mistake was ____."

4 · EXPLAIN IT · CHOOSE A FRAME

☐ The mistake was ____ because ____.
☐ The picture shows ____.
☐ I will check ____ next time.

I said it ☐ · I wrote it ☐

WORKSHOP 1 2 3

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Unit 1 review tic-tac-toe

19 D

DIRECTIONS · Pick the squares you most need. Three in a row, then try another line.

CALCULATE −2 m → +3 m → −1 m. Distance and displacement?	**DRAW** An x–t graph: away steadily, stop, come back faster.	**EXPLAIN** Why does a 3–4–5 trip give a 5 m displacement, not 7 m?
CALCULATE 0 → 12 m/s in 4 s, then how far (steady a)?	**CHALLENGE** ⋆ 3 kg cart: 9 N right, 3 N left, from rest for 2 s. Find v and Δx.	**DRAW** FBD of a book on a ramp at rest.
EXPLAIN Why don't 3rd-law pairs cancel?	**REAL WORLD** Why does a passenger lurch forward when a bus brakes?	**CALCULATE** A 40 N sign, one rope 15 N. The other rope?

Three in a row: _____ · The square that taught me most: _____

WORKSHOP 1 2 3

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Comic: fix a Unit 1 mistake

19 CS

YOUR READER
A classmate who keeps making one mistake. Pick one from today's review.

Stick figures are fine. The physics gets checked.

MUST-HAVES · check it off, then color its circle

- ☐ ◯ **PENCIL** — The mistake, crossed out with one line
- ☐ ◌ **PURPLE** — A velocity arrow (or an orange force arrow)
- ☐ ◌ **G/R/Y** — Green, red or yellow to show speeding up, slowing down or steady

1 · THE MISTAKE *A character says or does the wrong physics.*

CAPTION _____

2 · THE FIX *A friend explains the correct idea.*

CAPTION _____

3 · THE PROOF *Show evidence: a diagram, a graph or a calculation.*

CAPTION _____

EXPLAIN IT · In 1–2 sentences, what physics does your comic show? Use: **any 3 of velocity · acceleration · net force · displacement · equilibrium**

WORKSHOP 1 2 3

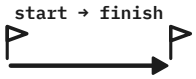
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Doodle Notes · displacement

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

Displacement · desplazamiento (ES) · deslocamento (PT)



TODAY'S CASE

A student walks 4 m right and 4 m left, then writes "my displacement is 8 m."

01 · SKETCH AND LABEL

Draw the route in pencil and mark the start and finish in blue. Circle the first mistaken claim.

02 · CHANGE THE PICTURE

Show the distance and displacement separately, with units. Use the diagram to explain one correction.

Annotate your sketches. Circle the mark that best shows displacement. Add a label in English or your home language.

SAY IT TO A PARTNER · THEN WRITE

The 8 m measures ___; the endpoints show …

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Frayer Model · displacement

I can explain a term by comparing an example with a nonexample.

Name: _____ My home language: _____

Displacement · desplazamiento (ES) · deslocamento (PT)

A · A total of 8 m walked guarantees 8 m of displacement.

B · Returning to the starting point gives zero displacement.

01 · DEFINITION

What does displacement mean in your own words? Use the notes or start in your home language.

02 · CHARACTERISTICS

What must be true? Label a small sketch to show one feature.

03 · EXAMPLE

Choose A or B as your example. Sketch it and explain which feature makes it fit.

04 · NONEXAMPLE

Explain why the other card does not fit. Change one detail so that it would.

SAY IT TO A PARTNER · THEN WRITE

I chose card ___ as my example because …

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Picture Dictionary · displacement

I can connect each word to a picture and use it to explain this case.

Name: _____ My home language: _____



mistake

error · erro

a step that is wrong

My word or label: _____



correct

corregir · corrigir

to fix it

My word or label: _____

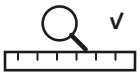


because

porque · porque

gives the reason

My word or label: _____

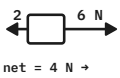


evidence

evidencia · evidência

the picture or numbers

My word or label: _____

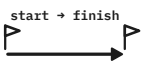


net force

fuerza neta · força resultante

all forces on one object combined

My word or label: _____



displacement

desplazamiento · deslocamento

change from the start to the finish

My word or label: _____

MATCH · WRITE THE ENGLISH WORD UNDER EACH PICTURE



_____



_____



_____

Use two words to explain the case: A student walks 4 m right and 4 m left, then writes "my displacement is 8 m."

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